

Question 1 (Start a new page)

Barker College

Limit 1999

Time 3 hrs.

MARKS

a) Find

3

i)  $\int \frac{4}{x^2 + 4} dx$

ii)  $\int \frac{4}{\sqrt{x^2 + 4}} dx$

iii)  $\int \frac{4x}{\sqrt{x^2 + 4}} dx$

b) Evaluate

8

i)  $\int_0^{\frac{\pi}{2}} e^x \cos x dx$

ii)  $\int_0^{\frac{\pi}{2}} \frac{d\theta}{2 + \cos \theta}$

c)

i) Find polynomials  $p(x), q(x)$  of degrees less than 2, such that  $(x+2)p(x) + (x^2+4)q(x) = 1$ .

4

ii) Hence evaluate  $\int_0^2 \frac{8dx}{(x+2)(x^2+4)}$ .

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**Question 2 (Start a new page)**

**MARKS**

a) If  $z = \frac{3+2i}{1-2i}$  then find

3

i)  $\bar{z}$

ii)  $\arg z$

b)

5

i) Express  $\sqrt{6i-8}$  in the form  $a+ib$  where  $a, b$  are elements of the set of reals.

ii) Hence solve  $2z^2 - (3+i)z + 2 = 0$  for  $z$ . Express your answer in the form  $a+ib$ .

c) Neatly sketch each of the following loci on separate Argand Diagrams.

4

i)  $\arg \frac{z+1}{z-i} = \frac{2\pi}{3}$

$$\arg \left( \frac{z+1}{z-i} \right) = \frac{\arg(z+1) - \arg(z-i)}{\arg(z+1) - \arg(z-i)}$$

ii)  $z\bar{z} = z + \bar{z}$

$$z\bar{z} - z - \bar{z} = 0$$

d)

3

i) Show on an Argand diagram the locus of  $z$  where  $|z-4-3i|=1$ .

ii) What are the least values of  $|z|$ .

3

**Question 3 (Start a new page)**

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a)

- i) Sketch  $y = f(x)$ , clearly labelling all essential features given that  $f(x) = x^3 - 4x$ .

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On separate diagrams sketch showing labelling all essential features

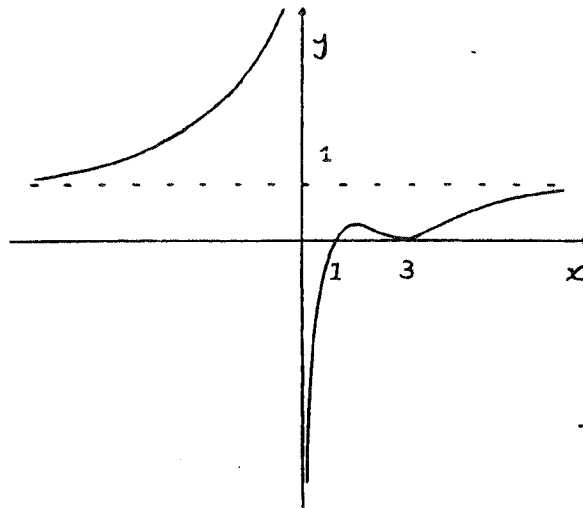
ii)  $y^2 = f(x)$

iii)  $y = f\left(\frac{1}{x}\right)$

iv)  $y = e^{f(x)}$

v)  $|y| = |f(x)|$

b)



5

The diagram above is of the derivative of  $y = f(x)$  . i.e. The curve has equation  $y = f'(x)$ .

- i) Sketch the function  $y = f''(x)$ .
- ii) On a separate diagram sketch a possible graph of  $y = f(x)$ .
- iii) Suggest a possible equation for  $y = f'(x)$  in terms of  $x$ .

4

**Question 4 (Start a new page)****MARKS**

a)

8

Show that the normal to the parabola  $x^2 = 4ay$  at the point  $P(2ap, ap^2)$  bisects the angle between the lines  $x = 2ap$  and  $SP$  where  $S$  is the focus of the parabola.

b)

7

- i) Sketch the hyperbola with equation  $\frac{x^2}{4} - \frac{y^2}{2} = 1$ , carefully labelling all essential features.
- ii) Show that the equation of the tangent to this hyperbola at  $P(2\sec\theta, \sqrt{2}\tan\theta)$  is given by  $\frac{x\sec\theta}{2} - \frac{y\tan\theta}{\sqrt{2}} = 1$ .
- iii) Hence prove that the area of the triangle bounded by this tangent and the asymptotes of the hyperbola is independent of the position of  $P$ .

**Question 5 (Start a new page)****MARKS**

a)

7

- i) Prove that the area enclosed by the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$  has a magnitude of  $\pi ab$ .
- ii) Find the volume of a mound with a circular base of equation  $x^2 + y^2 = 4$  which has semi-elliptical cross-sections parallel to the  $y$  axis, where the ratio of the major axis : minor axis = 2 : 1. The height of each cross-section is the length of the semi-minor axis.

b)

8

- i) Sketch the curve  $y = x^2(x^2 - 1)$  shading the region bounded by the curve and the  $x$ -axis.
- ii) Find the volume of the solid formed when this shaded area in part i) is rotated about the  $y$ -axis.
- iii) What is the volume of the solid formed when the area encompassed by the relation  $y^2 = x^8 - 2x^6 + x^4$  is rotated about the  $y$ -axis?

6.

**Question 6 (Start a new page)**

**MARKS**

- a) Show that  $1+i$  is a root of the polynomial  $P(x) = x^3 + x^2 - 4x + 6$  and hence completely factorize  $P(x)$  over the field of complex numbers. 3
- b) 4
- i) If the polynomial  $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$  has roots of the form  $a+ib$  and  $a-2ib$  where  $a, b$  are real, find the values of  $a$  and  $b$ .
- ii) Find all the zeros of  $P(x)$ .
- iii) Express  $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$  as a product of two quadratic factors with rational coefficients.
- c) 5
- i) Prove that if the polynomial  $P(x)$  has a root  $\alpha$  of multiplicity  $m$  then  $P'(x)$  has a root  $\alpha$  of multiplicity  $m-1$ .
- ii) Given that the polynomial  $P(x) = x^4 + x^3 - 3x^2 - 5x - 2$  has a root of multiplicity 3, find all the roots of  $P(x)$ .
- d) If  $\alpha, \beta, \gamma$  are the roots of the equation  $x^3 + qx + r = 0$ , prove that  $(\beta - \gamma)^2 + (\alpha - \beta)^2 + (\alpha - \gamma)^2 = -6q$ . 3

## Question 7 (Start a new page)

MAR!

a) If  $I_n = \int_0^1 \frac{x^n}{\sqrt{x+1}} dx$  where  $n = 0, 1, 2, 3, \dots$

7

i) Show that  $x^{n-1} \sqrt{x+1} = \frac{x^n}{\sqrt{x+1}} + \frac{x^{n-1}}{\sqrt{x+1}}.$

ii) Show that  $(2n+1)I_n = 2\sqrt{2} - 2nI_{n-1}$  for  $n = 1, 2, 3, \dots$

iii) Evaluate  $\int_0^1 \frac{x^2}{\sqrt{x+1}} dx.$

b)

8

i) Sketch on an argand diagram the roots of  $z^5 - 1 = 0.$

ii) Show that  $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}.$

iii) Hence or otherwise find the exact values of  $\cos \frac{2\pi}{5}$  and  $\cos \frac{\pi}{5}.$

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**Question 8 (Start a new page)****MARKS**

- a) Prove that if the opposite angles of a quadrilateral are supplementary then the quadrilateral must be cyclic.

**4**

b)

**5**

i) Show that  $\tan^{-1} a + \tan^{-1} b = \tan^{-1} \frac{a+b}{1-ab}$ .

ii) Simplify  $\tan^{-1} a + \tan^{-1} b + \tan^{-1} c$ .

iii) If the equation  $x^3 - 2x^2 + 3x + 4 = 0$  has roots  $\alpha, \beta, \gamma$  show that  $\tan^{-1} \alpha + \tan^{-1} \beta + \tan^{-1} \gamma = \frac{\pi}{4}$ .

c)

**6**

i) Show that  $f(x) = \frac{\sec x + \tan x}{2\sec x + 3\tan x}$  is a decreasing function in term of  $x$  for the domain  $0 < x < \frac{\pi}{2}$ .

ii) Deduce that  $\frac{\pi}{28} > \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sec x + \tan x}{2\sec x + 3\tan x} dx > (\sqrt{2} - 1) \frac{\pi}{12}$ .

**END OF PAPER**



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UNIT TRIAL 1999

$$(a) (i) \int \frac{4}{x^2+4} dx = 4 \cdot \frac{1}{2} \tan^{-1} \frac{x}{2} + C$$

$$= \underline{\underline{2 \tan^{-1} \frac{x}{2} + C}}$$

$$(ii) \int \frac{4}{\sqrt{x^2+4}} dx = \underline{\underline{4 \log_e (x + \sqrt{x^2+4}) + C}}$$

$$(iii) \int \frac{4x}{\sqrt{x^2+4}} = \frac{2(x^2+4)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= 4 \sqrt{x^2+4} + C$$

$$(b) (i) \int_0^{\frac{\pi}{2}} e^x \cos x dx = [e^x \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^x \sin x dx$$

$$= e^{\frac{\pi}{2}} - \left\{ [-e^x \cos x]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} e^x \cos x dx \right\}$$

$$2 \int_0^{\frac{\pi}{2}} e^x \cos x dx = e^{\frac{\pi}{2}} - 1$$

$$\therefore \int_0^{\frac{\pi}{2}} e^x \cos x dx = \underline{\underline{\frac{e^{\frac{\pi}{2}} - 1}{2}}}$$

$$(ii) \int_0^{\frac{\pi}{2}} \frac{du}{2 + \cos \theta} = \int_0^1 \frac{2}{2 + \frac{1-u^2}{1+u^2}} \cdot \frac{du}{u^2+1}$$

$$= \int_0^1 \frac{2}{2u^2 + 2 + 1 - u^2} du$$

$$= \int_0^1 \frac{2}{u^2 + 3} du$$

$$= \left[ \frac{2}{\sqrt{3}} \tan^{-1} \frac{u}{\sqrt{3}} \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}}$$

$$= \underline{\underline{\frac{\pi}{3\sqrt{3}} \text{ or } \frac{\sqrt{3}\pi}{9}}}$$

Let  $u = \tan \frac{\theta}{2}$   
 $du = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$   
 $d\theta = \frac{2 du}{u^2 + 1}$

$$(c) (x+2)p(x) + (x^2+4)q(x) = 1$$

$$\Rightarrow \frac{p(x)}{x^2+4} + \frac{q(x)}{x+2} = \frac{1}{(x+2)(x^2+4)}$$

Let  $p(x) = bx + c$  and  $q(x) = a$

$$(a+b)x^2 + (2b+c)x + 2a$$

$$a=1, b=-1, c=2$$

$$\int_0^2 \frac{1}{(x+2)(x^2+4)} dx = \int_0^2 \frac{1}{(x+2)(x^2+4)} dx$$

02

$$i) \quad i) \quad \bar{Z} = \frac{3+2i}{1-2i} \times \frac{1+2i}{1+2i}$$

$$= \frac{-1+8i}{5}$$

$$\therefore \bar{Z} = \underline{\underline{\frac{-1-8i}{5}}}$$

$$(ii) \quad \arg z = -\tan^{-1} 8 + \pi$$

$$ii) \quad \text{let } (a+ib)^2 = 6i-8$$

$$a^2 - b^2 + 2abi = 6i - 8$$

$$a^2 - b^2 = -8, \quad 2ab = 6 \Rightarrow ab = 3$$

$$a^2 - \frac{9}{a^2} = -8$$

$$\text{or } a^4 + 8a^2 - 9 = 0$$

$$(a^2 - 1)(a^2 + 9) = 0, \quad a \in \mathbb{R} \quad (i)$$

$$\therefore a = \pm 1, \quad b = \pm 3i$$

$$\therefore \sqrt{6i-8} = \pm(1+3i)$$

$$(ii) \quad 2z^2 - (3+i)z + 2 = 0$$

$$z = \frac{3+i \pm \sqrt{(3+i)^2 - 16}}{4}$$

$$= \frac{3+i \pm \sqrt{6i-8}}{4}$$

$$= \underline{\underline{\frac{3+i \pm (1+3i)}{4}}}$$

$$z = 1+i \pm \underline{\underline{\frac{1-i}{2}}}$$

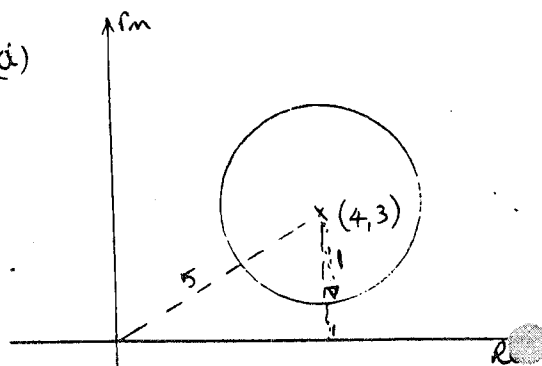
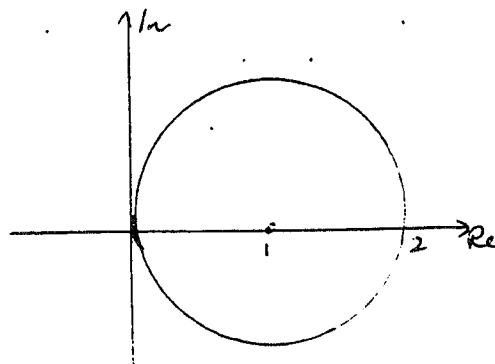
$$C(ii) \quad z = a+ib$$

$$\therefore (a+ib)(a-ib) = a+ib + a-ib$$

$$a^2 - b^2 = 2a$$

$$a^2 - 2a - b^2 = 0$$

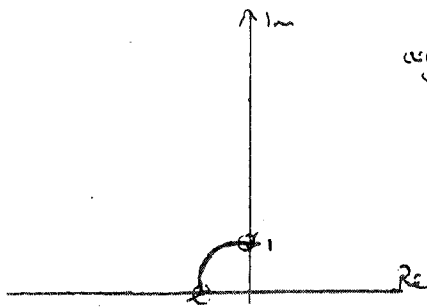
$$(a-1)^2 - b^2 = 1$$



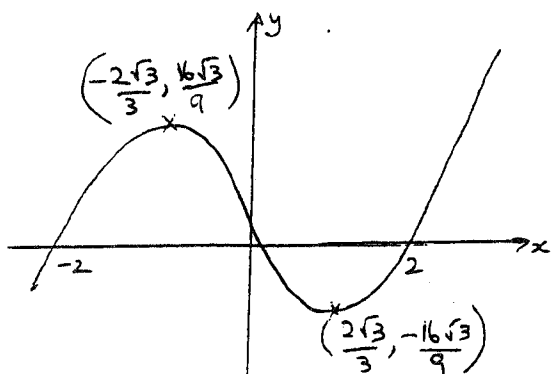
Least value of  $|z| = 4$

ii) i)

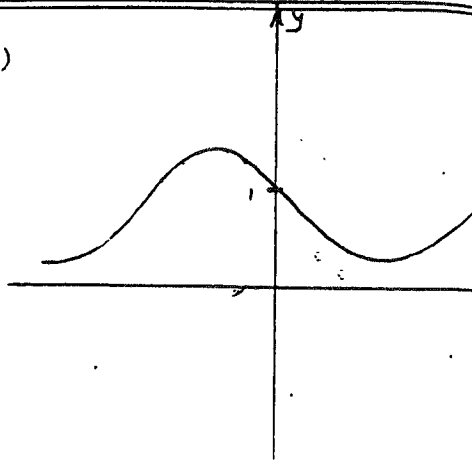
$$\arg(z+1) > \arg(z-i)$$



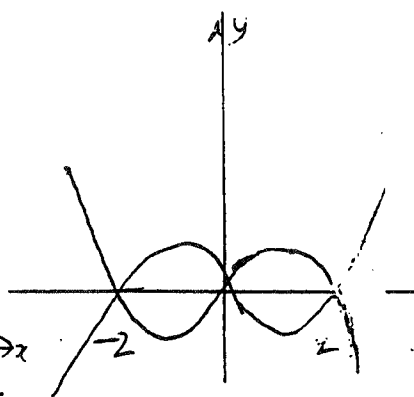
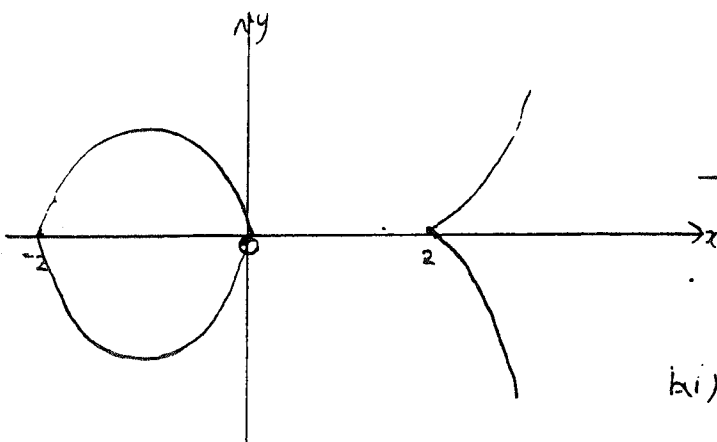
3. i)  $f(x) = x^3 - 4x$   
 $f(x) = x(x-2)(x+2)$   
 $f'(x) = 3x^2 - 4$   
 $= (\sqrt{3}x - 2)(\sqrt{3}x + 2)$



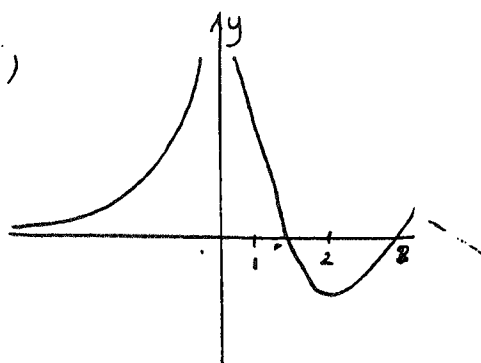
(iv)



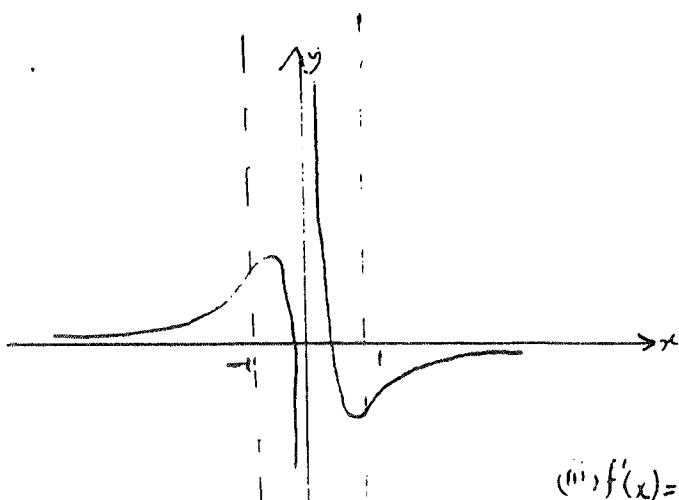
(ii)



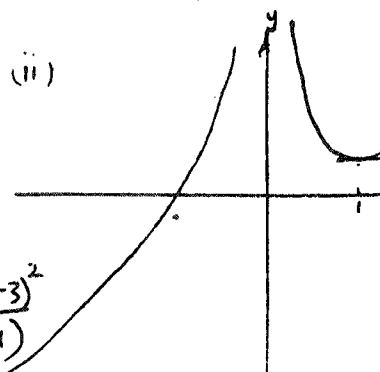
(vi)



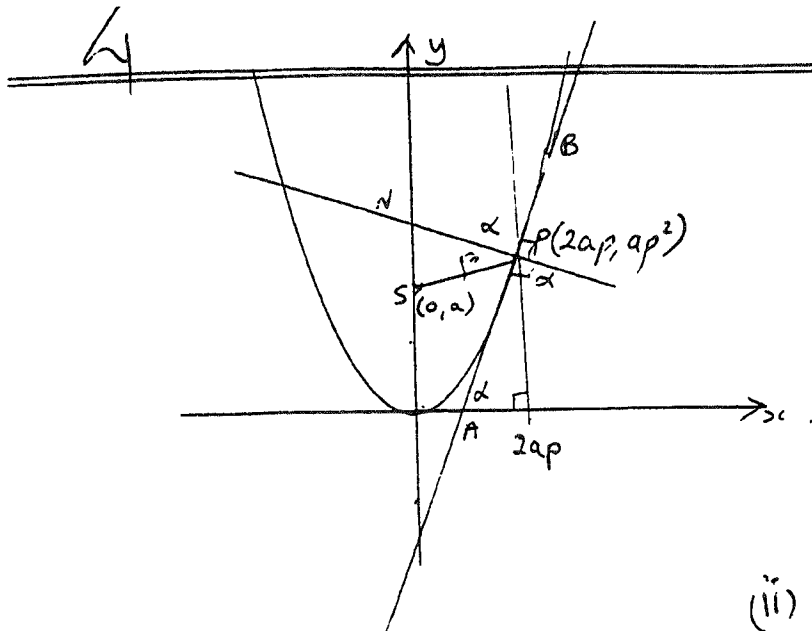
(iii)



(ii)



(iii)  $f'(x) = \frac{(x-1)(x-3)^2}{x(x^2+1)}$



$$M_{AB} = \tan \alpha = p$$

$$M_{SP} = \frac{ap^2 - a}{2ap}$$

$$= \frac{p^2 - 1}{2p}$$

$$M_{PN} = -\frac{1}{p}$$

$$\tan \beta = \left| \frac{\frac{p^2 - 1}{2p} + \frac{1}{p}}{1 - \frac{1}{p} \left( \frac{p^2 - 1}{2p} \right)} \right|$$

$$= |p|$$

$$\text{As } \alpha, \beta < \frac{\pi}{2}$$

$$\text{and } \tan \alpha = \tan \beta \quad \underline{\underline{\alpha = \beta}}$$

angle is bisected

$$(ii) \quad \frac{2x - 2y}{4} \cdot \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{x}{2y}$$

$$\text{When } x = 2 \sec \theta \quad y = \sqrt{2} \tan \theta$$

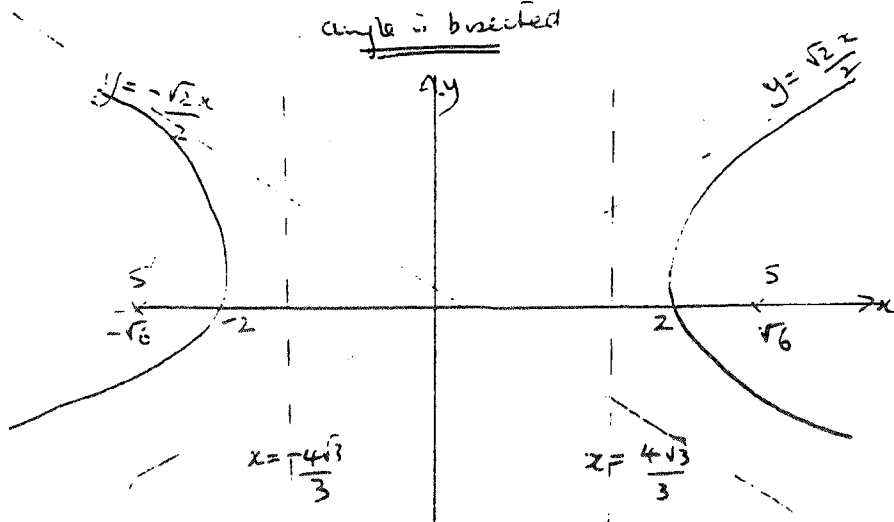
$$\therefore \frac{dy}{dx} = \frac{\sec \theta}{\sqrt{2} \tan \theta}$$

$$\therefore y - \sqrt{2} \tan \theta = \frac{\sec \theta}{\sqrt{2} \tan \theta}$$

$$\therefore x \sec \theta - \sqrt{2} \tan \theta \cdot y = 2 \sec^2 \theta - 2 \tan^2 \theta = 2$$

$$\therefore \frac{x \sec \theta}{2} - \frac{y \tan \theta}{\sqrt{2}} = 1$$

is equation of tangent

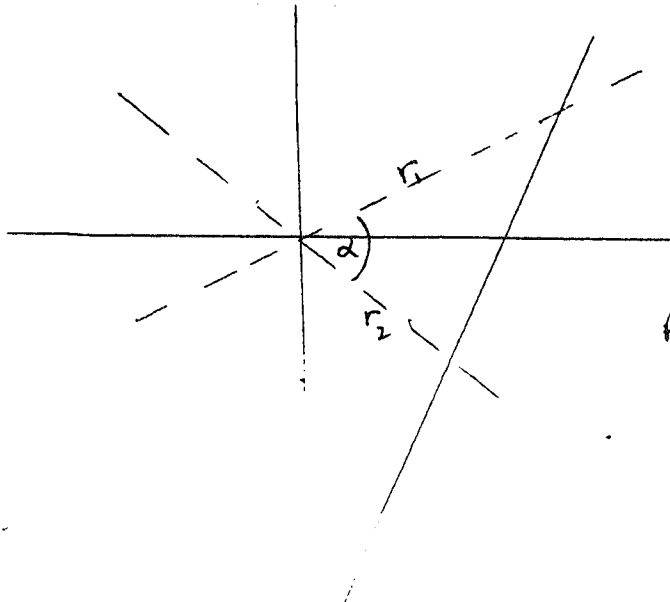


(iii) Find the points of intersection at asymptotes and tangent

$$\begin{aligned} \therefore y &= \frac{\sqrt{2}x}{2} \\ \therefore \frac{x \sec \theta}{2} - \frac{x \tan \theta}{2} &= 1 \end{aligned}$$

$$\therefore x = \frac{2}{\sec \theta - \tan \theta}, \quad y = \frac{\sqrt{2}}{\sec \theta - \tan \theta}$$

$$\begin{aligned} \text{for } y &= -\frac{\sqrt{2}x}{2} \\ \frac{x \sec \theta}{2} + \frac{x \tan \theta}{2} &= 1 \\ \therefore x &= \frac{2}{\sec \theta + \tan \theta}, \quad y = \sec \theta - \tan \theta \end{aligned}$$



Area  $\Delta = \frac{1}{2} r_1 r_2 \sin \alpha$   
where  $\alpha$  is const.

$$\text{Now } r_1^2 = \left( \frac{2}{\sec \theta - \tan \theta} \right)^2 + \left( \frac{\sqrt{2}}{\sec \theta - \tan \theta} \right)^2$$

$$\therefore r_1 = \frac{\sqrt{6}}{\sec \theta - \tan \theta}$$

$$\text{and } r_2 = \frac{\sqrt{6}}{\sec \theta + \tan \theta}$$

$$\text{Area} = \frac{1}{2} \frac{\sqrt{6}}{\sec \theta - \tan \theta} \cdot \frac{\sqrt{6}}{\sec \theta + \tan \theta} \cdot \sin \alpha$$

$$= \frac{3}{2} \frac{1}{\sec^2 \theta - \tan^2 \theta} \sin \alpha$$

$$= \frac{3}{2} \sin \alpha \text{ which is constant}$$

$\therefore$  Area is independent of the position of P

5

P14

$$(i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

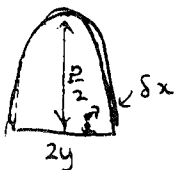
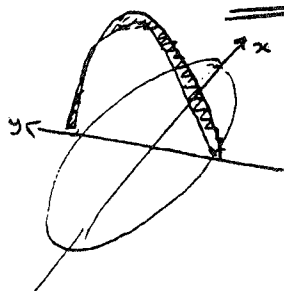
$$\therefore y = \frac{b}{a} \sqrt{a^2 - x^2}$$

$$\therefore \text{Area} = 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$$

$$= 4 \cdot \frac{b}{a} \cdot \frac{\pi a^2}{4}$$

$$= \pi ab$$

(ii)



$$V = \frac{1}{2} \pi a \cdot \frac{b}{2} \delta x$$

$$= \frac{\pi a^2}{8} \delta x$$

$$= \frac{\pi y^2}{2} \delta x$$

$$\therefore V = 2 \int_0^2 \frac{\pi y^2}{2} dx$$

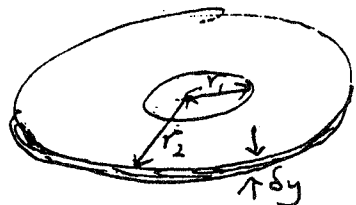
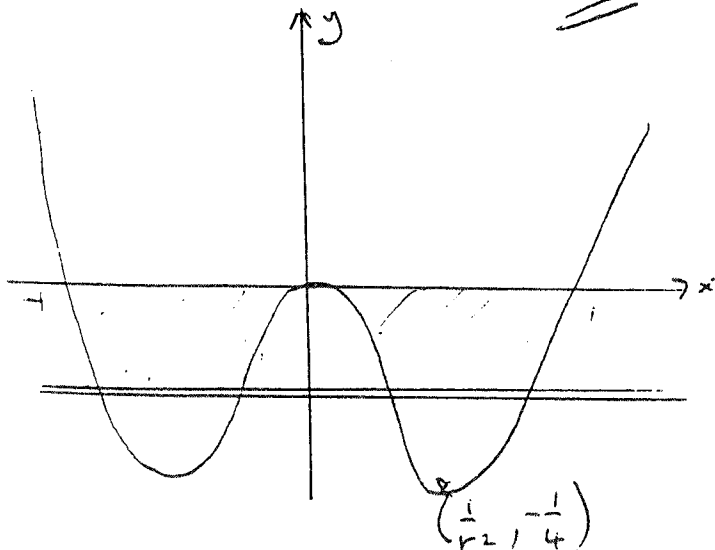
$$= \pi \int_0^2 4 - x^2 dx$$

$$= \pi \left[ 4x - \frac{x^3}{3} \right]_0^2$$

$$= \pi \left[ \frac{16}{3} \right]$$

$$= \frac{16\pi}{3}$$

(b)



$$\text{Volume of slice} = \pi (r_2^2 - r_1^2) \delta y$$

$$\text{As } x^4 - x^2 - y = 0$$

$$\therefore x^2 = \frac{1 \pm \sqrt{1+4y}}{2}$$

$$\therefore r_2^2 = \frac{1 + \sqrt{1+4y}}{2}$$

$$\text{and } r_1^2 = \frac{1 - \sqrt{1+4y}}{2}$$

$$\therefore r_2^2 - r_1^2 = \sqrt{1+4y}$$

$$\therefore V = \pi \int_{-\frac{1}{4}}^0 \sqrt{1+4y} dy$$

$$= \pi \left[ \frac{2}{3 \times 4} (1+4y)^{\frac{3}{2}} \right]_{-\frac{1}{4}}^0$$

$$= \frac{\pi}{6}$$

$$\text{As } y = x^2(x^2 - 1)$$

$$\text{Volume} = \frac{2 \times \pi}{6}$$

$$= \frac{\pi}{3} \text{ units}^3$$

p15

(a)  $1+i$ 

$$(1+i)^2 = 1 + i^2 + 2i$$

$$= 2i$$

$$(1+i)^3 = (1+i)2i$$

$$= 2i - 2$$

$$P(1+i) = 2i - 2 + 2i - 4(1+i) + 6$$

$$= 0$$

$\therefore 1+i$  is a root.

$\therefore 1-i$  is a root

$x^2 - 2x + 2$  is a factor

$$\therefore P(x) = (x - (1+i))(x - (1-i))(x+3)$$

$$= (x^2 - 2x + 2)(x+3)$$

(b) (i)

$$\text{Sum of roots} = 4 = 4a$$

$$\therefore a = 1$$

$$\text{Product of roots} = 10 \therefore (a^2 - b^2)(a^2 + 4b^2) = 10$$

$$a = 1, \quad (1 - b^2)(1 + 4b^2) = 0$$

$$\therefore (1 - b^2 - 4b^2) = 0$$

$$\therefore b^2 = \frac{9}{4}$$

$$\therefore b = \pm \frac{3}{2}$$

(ii)

$$\therefore \text{Roots are } 1 + \frac{3i}{2}, 1 - \frac{3i}{2}, 1 - 3i, 1 + 3i$$

(iii)

$$P(x) = (x^2 - 2x - \frac{5}{4})(x^2 - 2x - 8)$$

(c) (i)

$$\text{Let } P(x) = (x - \alpha)^m Q(x) \text{ where } x - \alpha \nmid Q(x)$$

$$\therefore P'(x) = m(x - \alpha)^{m-1} Q(x) + (x - \alpha)^m Q'(x)$$

$$= (x - \alpha)^{m-1} [m Q(x) + (x - \alpha) Q'(x)]$$

$$\text{now } x - \alpha \nmid (x - \alpha) Q'(x) \text{ but not } m Q(x)$$

$$\therefore x - \alpha \mid [m Q(x) + (x - \alpha) Q'(x)]$$

$\therefore$  root  $\alpha$  has multiplicity  $m$  in  $P(x)$

multiplicity  $m-1$  in  $P'(x)$

(ii)

$$P(x) = x^4 + x^3 - 3x^2 - 5x - 2$$

$$P'(x) = 4x^3 + 3x^2 - 6x - 5$$

$$P''(x) = 12x^2 + 6x - 6$$

$$= 6(2x^2 + x - 1)$$

$$= \underline{6(2x-1)(x+1)}$$

$$\therefore P''(-1) = 0 \quad P''\left(\frac{1}{2}\right) = 0 \quad P'(-1) = 0$$

$$\text{At } P(-1) = 1 - 1 - 3 + 5 - 2$$

$$= 0$$

$$\therefore (x+1) \text{ is a root of } P(x)$$

$$\therefore P(x) = (x+1)^3(x-2)$$

(d)

$$\text{given roots } x = -1, 2$$

$$2(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 2[(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)] - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 2(\alpha + \beta + \gamma)^2 - 6(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 2 \cdot 0^2 - 6q$$

$$= \underline{\underline{-6q}}$$



$$\begin{aligned}
 (i) \quad \frac{x^n}{\sqrt{x+1}} + \frac{x^{n-1}}{\sqrt{x+1}} &= \frac{x^n \sqrt{x+1} + x^{n-1} \sqrt{x+1}}{x+1} \\
 &= \sqrt{x+1} \left[ \frac{x^{n-1}(x+1)}{x+1} \right] \\
 &= \sqrt{x+1} \cdot x^{n-1}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad I_n &= \int_0^1 \frac{x^n}{\sqrt{x+2}} dx \\
 &= \left[ x^n \cdot 2\sqrt{x+1} \right]_0^1 - \int_0^1 n x^{n-1} \cdot 2\sqrt{x+1} dx \\
 &= 2\sqrt{2} - n \int_0^1 x^{n-1} \cdot 2\sqrt{x+1} dx \\
 &= 2\sqrt{2} - 2n \int_0^1 \frac{x^n}{\sqrt{x+1}} + \frac{x^{n-1}}{\sqrt{x+1}} dx
 \end{aligned}$$

$$I_n(2n+1) = 2\sqrt{2} - 2nI_{n-1}$$

$$\begin{aligned}
 (iii) \quad I_0 &= \int_0^1 \frac{1}{\sqrt{x+1}} dx \\
 &= \left[ 2\sqrt{x+1} \right]_0^1 \\
 &= 2(\sqrt{2}-1)
 \end{aligned}$$

$$\begin{aligned}
 I_1(3) &= 2\sqrt{2} - 2I_0 \\
 &= 2\sqrt{2} - 2(2\sqrt{2}-2) \\
 &= 4 - 2\sqrt{2} \\
 I_1 &= \frac{4}{3} - \frac{2}{3}\sqrt{2}
 \end{aligned}$$

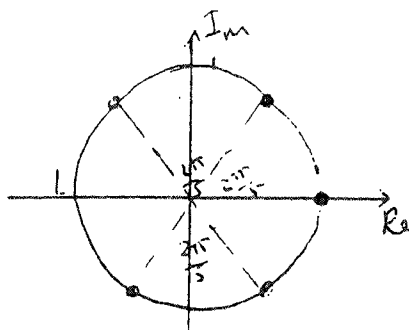
$$5I_2 = 2\sqrt{2} - 4I_1 \quad \text{and} \quad 3I_1 = 2\sqrt{2} - 2I_0$$

$$\begin{aligned}
 \therefore I_1 &= \frac{2}{3}\sqrt{2} - \frac{4}{3}(\sqrt{2}-1) \\
 &= \frac{4}{3} - \frac{2}{3}\sqrt{2}
 \end{aligned}$$

$$I_2 = \frac{2}{5}\sqrt{2} - \frac{4}{5} \left( \frac{4}{3} - \frac{2}{3}\sqrt{2} \right)$$

$$\begin{aligned}
 &= \frac{16\sqrt{2}}{15} - \frac{16}{15} = \frac{2}{15}(7\sqrt{2}-5)
 \end{aligned}$$

(i)



$$\begin{aligned}
 (ii) \quad \sum \text{roots} &= 1 + \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} + \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5} + \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} \\
 &= 1 + 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5}
 \end{aligned}$$

$$\text{But } \sum \text{roots} = 0$$

$$1 + 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} = 0$$

$$\begin{aligned}
 (iii) \quad \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} &= 2\cos^2 \frac{2\pi}{5} - 1
 \end{aligned}$$

$$\text{Let } \omega = \cos \frac{2\pi}{5}$$

$$\omega + 2\omega^2 - 1 = -\frac{1}{2}$$

$$4\omega^2 + 2\omega - 1 = 0$$

$$\omega = \frac{-2 \pm \sqrt{20}}{8}$$

$$= \frac{-1 \pm \sqrt{5}}{4}$$

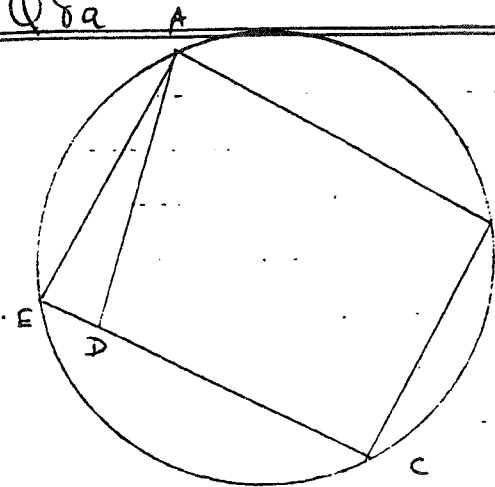
$$\text{But } \cos \frac{2\pi}{5} > 0$$

$$\therefore \cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$$

$$\begin{aligned}
 \cos \frac{4\pi}{5} &= 2 \cdot \left( \frac{\sqrt{5}-1}{4} \right)^2 - 1 \\
 &= \frac{-1-\sqrt{5}}{4}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \cos \frac{\pi}{5} &= \frac{1+\sqrt{5}}{4}
 \end{aligned}$$

Q8a



Draw a circle through ABC and assume it will not pass through D. (ie ABCD is not cyclic!)

Produce CD to E a point on the circle

Now  $\angle ADC + \angle ABC = 180^\circ$  (opp angle quad)

Also  $\angle AEC + \angle ABC = 180^\circ$  (opp angle of cyclic quad)

$$\therefore \angle AEC = \angle ADC$$

But  $\angle AEC, \angle ADC$  are corresponding angles

$$\therefore AE \parallel DA$$

But this is not possible as A is common to both

$\therefore$  Assumption is incorrect  $\therefore$  ABCD is cyclic

b(i)

$$\tan^{-1}(\tan(\tan^{-1}a - \tan^{-1}b)) = \tan^{-1}\left[\frac{\tan(\tan^{-1}a) - \tan(\tan^{-1}b)}{1 + \tan(\tan^{-1}a)\tan(\tan^{-1}b)}\right]$$

(ii)

$$= \tan^{-1}\left(\frac{a-b}{1+ab}\right)$$

$$\tan^{-1}a + \tan^{-1}b + \tan^{-1}c = \tan^{-1}\left[\frac{\frac{a+b}{1-ab} + c}{1 - \frac{a+b}{1-ab} \cdot c}\right]$$

$$= \tan^{-1}\left[\frac{a+b+c-abc}{1-(ab+ac+bc)}\right]$$

(iii)

$$\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma = \tan^{-1}\left(\frac{2-4}{1-3}\right) = \tan^{-1}\left(\frac{-2}{-2}\right)$$

$$= \frac{\pi}{4}$$

$$8 \text{ (i)} \quad f(x) = \frac{\sec x + \tan x}{2 \sec x + 3 \tan x}$$

$$(2 \sec x \tan x + 2 \sec^2 x)$$

$$f'(x) = \frac{(2 \sec x + 3 \tan x)(\sec x \tan x + \sec^2 x) - (\sec x + \tan x)(2 \sec^2 x + 3 \sec x \tan x)}{(2 \sec x + 3 \tan x)^2}$$

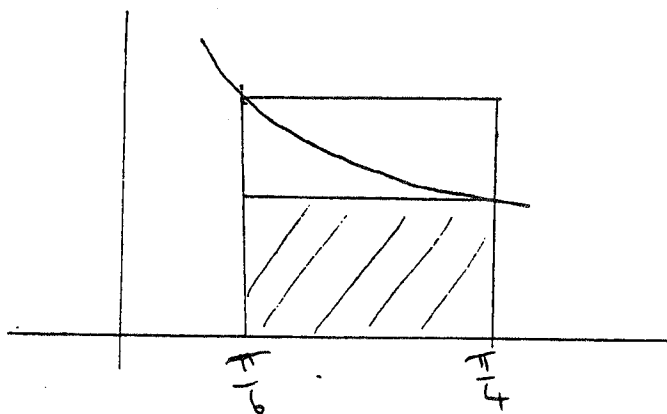
$$= \frac{-\sec x (\tan^2 x - \sec^2 x)}{(2 \sec x + 3 \tan x)^2}$$

$$= \frac{-\sec x}{(2 \sec x + 3 \tan x)^2}$$

now  $\sec x > 0$   
for  $0 \leq x < \frac{\pi}{2}$

$$\therefore \underline{f'(x) < 0} \text{ for } 0 \leq x < \frac{\pi}{2}$$

(ii)



$$A_{\text{upper}} = \left( \frac{\pi}{4} - \frac{\pi}{6} \right) \cdot \frac{\sec \frac{\pi}{6} + \tan \frac{\pi}{6}}{2 \sec \frac{\pi}{6} + 3 \tan \frac{\pi}{6}}$$

$$= \frac{\pi}{12} \left[ \frac{\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{2 \cdot \frac{2}{\sqrt{3}} + 3 \cdot \frac{1}{\sqrt{3}}} \right]$$

$$= \frac{\pi}{28}$$

$$A_{\text{lower}} = \left( \frac{\pi}{4} - \frac{\pi}{6} \right) \cdot \frac{\sec \frac{\pi}{4} + \tan \frac{\pi}{4}}{2 \sec \frac{\pi}{4} + 3 \tan \frac{\pi}{4}}$$

$$= \frac{\pi}{12} \frac{\sqrt{2} + 1}{2\sqrt{2} + 3} \times \frac{2\sqrt{2} - 3}{2\sqrt{2} - 3}$$

$$= \frac{\pi}{12} \cdot \frac{4 - 3 + \sqrt{2}}{3 - 9}$$

$$= \frac{\pi}{12} (\sqrt{2} - 1)$$

$$\therefore \frac{\pi}{28} > \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{\sec x + \tan x}{2 \sec x + 3 \tan x} dx > (\sqrt{2} - 1) \cdot \frac{\pi}{12}$$